Group 12

DRL Tracker
Use-Case Specification

Version 1.0

Revision History

Date	Version	Description	Author
01/11/2026	1.0	Completed basic content	Tú, Vĩ

©Group 12, 2026

Table of Contents

1. Use-case Model **4**

2. Use-case Specifications **4**

2.1 Use-case UC01: Discover & Register for Activities 4

2.2 Use-case UC02: Manage Evidence 5

2.3 Use-case UC03: Track Training Points 5

2.4 Use-case UC04: Submit Appeal 6

2.5 Use-case UC05: Interact with AI Assistant 6

2.6 Use-case UC06: Manage Activity Lifecycle 7

2.7 Use-case UC07: Manage Participant List 8

2.8 Use-case UC08: Verify Evidence 8

2.9 Use-case UC09: Process & Forward Appeals 9

2.10 Use-case UC10: System Audit Trail 9

2.11 Use-case UC11: Handle Exception Cases 10

2.12 Use-case UC12: Aggregate Points & Propose (Faculty) 11

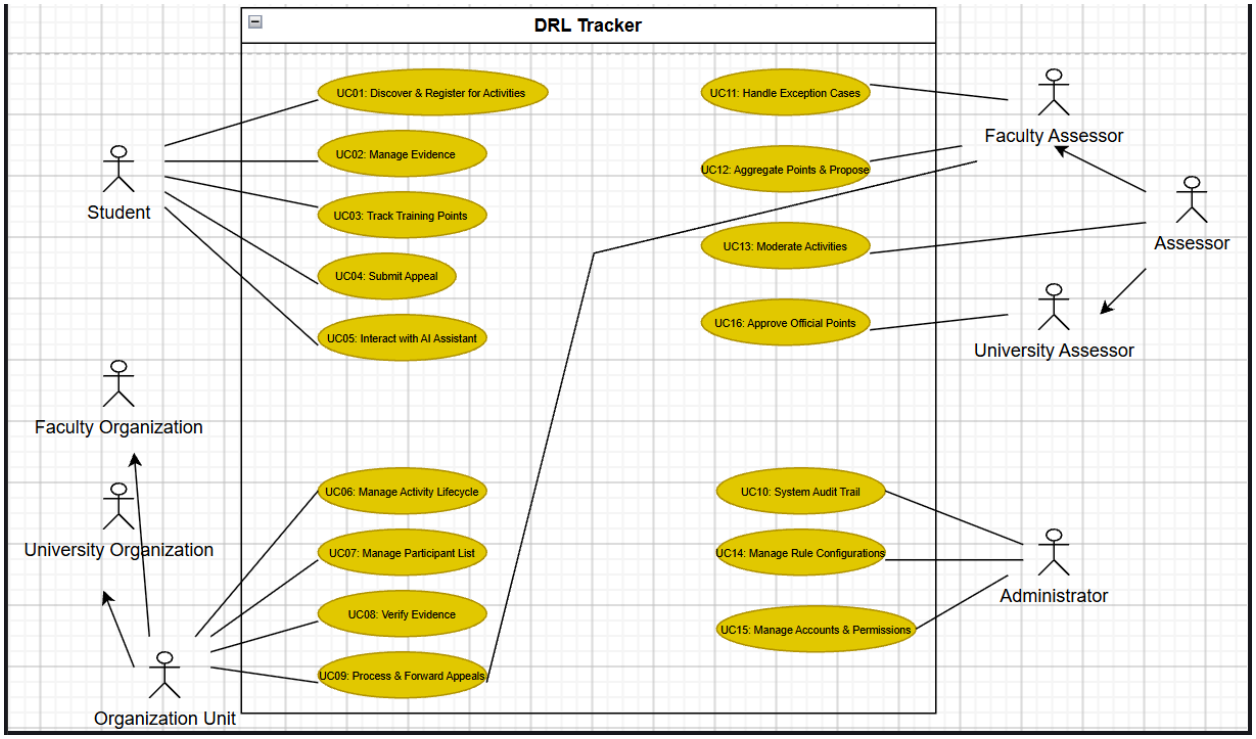
2.13 Use-case UC13: Moderate Activities 11

2.14 Use-case UC14: Manage Rule Configurations 12

2.15 Use-case UC15: Manage Accounts & Permissions 13

2.16 Use-case UC16: Approve Official Points (University) 13

1. Use-case Model



2. Use-case Specifications

2.1 Use-case UC01: Discover & Register for Activities

Use case Name	Discover & Register for Activities.
Brief description	This use case allows a Student to search, view details, and register for extracurricular activities.
Actors	Student.
Basic Flow	1. At the Dashboard, the Student navigates to the activity discovery section. 2. The Student inputs search keywords or applies filters (by unit, criteria). 3. The system returns matching activities. 4. The Student clicks on an activity to view details (time, location, capacity, deadline). 5. The Student clicks the "Register" button. 6. The system verifies the capacity and registration deadline. 7. The system saves the registration and updates the participant list. 8. The system displays a success message.
Alternative Flows	**Alternative flow 1: Registration full or expired** 1. At step 6 of the basic flow, if the activity is full or past the deadline, the system disables the "Register" button or shows an error message. 2. The system prompts the Student to choose another activity. **Alternative Flow 2: Cancel Activity Registration** 1. The Student navigates to their registered activities.

	2. The Student selects an activity and clicks "Cancel Registration". 3. The system checks whether the cancellation request is submitted at least one day before the scheduled activity date. 4. If the condition is satisfied, the system cancels the registration and updates the participant list. 5. The system displays a cancellation confirmation message. **Alternative Flow 3: Cancellation Deadline Passed** 1. At step 3 of Alternative Flow 2, if the cancellation request is submitted less than one day before the scheduled activity date, the system rejects the request. 2. The system displays a message informing the Student that the cancellation deadline has passed.
Pre-conditions	The Student is logged into the system.
Post-conditions	The Student is successfully registered for the activity, or their registration is successfully cancelled.

2.2 Use-case UC02: Manage Evidence

Use case Name	Manage Evidence.
Brief description	This use case allows a Student to upload and track participation evidence for registered activities.
Actors	Student.
Basic Flow	1. The Student goes to the evidence management section. 2. The Student selects a specific activity. 3. The system displays the upload form. 4. The Student uploads an image/document and adds an optional description. 5. The Student clicks "Submit". 6. The system verifies the file format and size. 7. The system saves the evidence with a "PENDING" status. 8. The system displays a success message.
Alternative Flows	**Alternative flow 1: Invalid file format or size** 1. At step 6 of the basic flow, if the file is invalid, the system displays an error message. 2. The system prompts the Student to upload a valid file. **Alternative Flow 2: Submit Additional Evidence** 1. The Student selects an evidence record with "REQUIRES ADDITIONAL EVIDENCE" status. 2. The system displays the additional information requested by the Organization or Assessment Officer. 3. The Student uploads the requested additional evidence. 4. The system validates the uploaded file. 5. The system updates the evidence status to "PENDING". 6. The system notifies the responsible reviewer.
Pre-conditions	The Student is logged in and has registered for the target activity.
Post-conditions	The evidence is securely stored and awaits verification from the Organizing Unit.

2.3 Use-case UC03: Track Training Points

Use case Name	Track Training Points.
Brief description	This use case enables a Student to view their aggregated points by semester, distinguishing between Maximum, Predicted, and Official Scores.
Actors	Student.
Basic Flow	1. The Student navigates to the training points dashboard. 2. The system retrieves the student's records for the current semester. 3. The system displays the Maximum Score based on active regulations. 4. The system calculates and displays the Predicted Score based on valid evidence. 5. The system displays the Official Score only if the University Assessment Officer has approved and finalized the student's training point results. Otherwise, the Official Score is not displayed. 6. The Student views the breakdown of points by criteria.
Alternative Flows	**Alternative flow 1: Score discrepancies found** 1. At step 6 of the basic flow, if the Student notices missing points, they can proceed to UC04 (Submit Appeal).
Pre-conditions	The Student is logged into the system.
Post-conditions	The Student is informed of their current training point status.

2.4 Use-case UC04: Submit Appeal

Use case Name	Submit Appeal.
Brief description	This use case allows a Student to submit a review request (complaint) for invalid evidence or score discrepancies.
Actors	Student.
Basic Flow	1. The Student selects the rejected evidence or specific score component. 2. The Student clicks the "Submit Appeal" button. 3. The system verifies that the 1-week deadline from status update has not passed. 4. The system verifies the appeal count is under the 5-time limit. 5. The Student enters a justification and uploads additional proof if needed. 6. The Student clicks "Submit". 7. The system determines the appropriate recipient based on the activity type: - For activities organized by an Organization, the appeal is sent to the corresponding Organization. - For exceptional activities outside the system, the appeal is sent to the Faculty level Assessment Officer. 8. The system records the appeal with a "PENDING" status and notifies the responsible party.
Alternative Flows	**Alternative flow 1: Limit exceeded or deadline passed** 1. At step 3 or 4, if the limit is reached or the deadline has passed, the system displays an error message. 2. The appeal submission is blocked.
Pre-conditions	The Student has participated in the target activity. The relevant evidence or assessment status has been updated. The Student has not exceeded the 5-time

	appeal limit.
Post-conditions	The appeal is recorded and sent to the appropriate responsible party (the corresponding Organization or the Faculty-level Assessment Officer) for processing.

2.5 Use-case UC05: Interact with AI Assistant

Use case Name	Interact with AI Assistant.
Brief description	A Student interacts with a chatbot for personalized activity recommendations based on missing criteria.
Actors	Student.
Basic Flow	1. The Student opens the AI Assistant chatbox. 2. The Student types a query regarding activity suggestions or point shortages. 3. The AI retrieves the Student's Predicted Score, missing criteria, preferences, and available schedule information. 4. The AI analyzes available upcoming activities based on the Student's preferences, suitable time slots, and missing training point criteria. 5. The AI responds with suitable activity recommendations.
Alternative Flows	**Alternative flow 1: Unauthorized action requested** 1. At step 2, if the Student asks the AI to modify a score or verify evidence, the AI refuses the request. 2. The AI explains that it only has read/consult permissions and cannot modify scores, verify evidence, or make assessment decisions.
Pre-conditions	The Student is logged in.
Post-conditions	The Student receives guidance without any modifications to actual system data.

2.6 Use-case UC06: Manage Activity Lifecycle

Use case Name	Manage Activity Lifecycle.
Brief description	This use case allows Organizations to create, edit, or cancel activities and submit them for moderation within their assigned management scope. Faculty-level Organizations manage faculty activities, while university-level Organizations manage university-level activities.
Actors	Organization (Faculty Level or University Level).
Basic Flow	1. The Organizing Unit clicks "Create Activity" on the Dashboard. 2. The Organizing Unit inputs details (name, description, time, location, capacity, criteria, deadline). 3. The Organizing Unit clicks "Submit for Approval". 4. The system validates required fields. 5. The system saves the activity with a "Pending Approval" status. 6. The system notifies the Assessor.
Alternative Flows	**Alternative flow 1: Cancel an activity** 1. The Organizing Unit selects an existing activity and clicks "Cancel". 2. The Organizing Unit inputs a reason. 3. The system updates the status to "Cancelled" and notifies registered students. **Alternative Flow 2: Edit Activity**

	1. The Organization selects an existing activity within its management scope. 2. The Organization clicks "Edit Activity". 3. The system displays the current activity information. 4. The Organization modifies the necessary details. 5. The Organization submits the changes. 6. The system validates the updated information. 7. The system saves the changes and updates the activity information. If the activity requires re-approval due to significant changes, the system updates its status to "Pending Approval" and notifies the responsible Assessment Officer.
Pre-conditions	The Organizing Unit has valid permissions and has management authority over the target activity.
Post-conditions	The activity is created (pending approval) or updated accordingly.

2.7 Use-case UC07: Manage Participant List

Use case Name	Manage Participant List.
Brief description	The Organizing Unit tracks registered students and finalizes the actual attendance list after the event.
Actors	Organizing Unit.
Basic Flow	1. The Organizing Unit selects a completed activity. 2. The system displays the list of registered students. 3. The Organizing Unit uploads or manually updates the actual attendance data. 4. The Organizing Unit clicks "Finalize List". 5. The system saves the final participant list.
Alternative Flows	**Alternative flow 1: Student missing from registration** 1. At step 3, if a student participated but is not on the registered list, the Organizing Unit can manually add the student's ID. 2. Resume from step 4.
Pre-conditions	The activity is in "In Progress" or "Completed" status.
Post-conditions	The final attendance list is locked for the activity.

2.8 Use-case UC08: Verify Evidence

Use case Name	Verify Evidence.
Brief description	The Organizing Unit cross-checks student evidence against the actual participant list and updates the status.
Actors	Organizing Unit.
Basic Flow	1. The Organizing Unit views the pending evidence list on the Dashboard. 2. The Organizing Unit selects an evidence file to review. 3. The Organizing Unit cross-checks the evidence with the finalized participant list. 4. The Organizing Unit clicks "Valid". 5. The system updates the evidence status to "VALID".
Alternative Flows	**Alternative flow 1: Invalid evidence** 1. At step 4, if the evidence is blurry or fake, the Organizing Unit clicks "Reject". 2. The system prompts for a rejection reason. 3. The Organizing Unit inputs the reason and saves.

	4. The system updates the status to "INVALID" and notifies the Student. **Alternative Flow 2: Request Additional Evidence** 1. At step 3 of the basic flow, if the evidence is insufficient or requires further clarification, the Organization clicks "Request Additional Evidence". 2. The Organization enters the reason and specifies the additional information required. 3. The system updates the evidence status to "REQUIRES ADDITIONAL EVIDENCE". 4. The system notifies the Student about the request. 5. The Student submits the additional evidence through UC02. 6. The system updates the evidence status to "PENDING" and makes it available for re-verification.
Pre-conditions	The Student has submitted evidence ("PENDING" status) and the attendance list is finalized.
Post-conditions	The evidence is marked as VALID, INVALID, or PENDING. The verification result is recorded and affects the student's Predicted Score when applicable.

2.9 Use-case UC09: Process & Forward Appeals

Use case Name	Process & Forward Appeals.
Brief description	Organizations to handle student complaints related to activities managed by their units. Complaints concerning exceptional activities outside the system are handled separately by Faculty-level Assessment Officers.
Actors	Organizing (Faculty Level or University Level). Faculty-level Assessment Officer.
Basic Flow	1. The Organization selects a pending appeal related to an activity managed by its unit. 2. The system displays the Student's justification, submitted evidence, and related activity information. 3. The Organization reviews the appeal and verifies the provided information. 4. The Organization decides whether the appeal is valid. 5. If the appeal is approved, the system updates the relevant evidence or assessment status accordingly. 6. If the appeal is rejected, the Organization enters a rejection reason. 7. The system records the decision and notifies the Student.
Alternative Flows	**Alternative flow 1: Request Additional Evidence** 1. At step 3 of the basic flow, if the provided information is insufficient, the Organization requests additional evidence. 2. The Organization enters the reason and requires information. 3. The system notifies the Student and updates the appeal status to "REQUIRES ADDITIONAL EVIDENCE". 4. The Student submits the requested information. 5. The appeal returns to the Organization for further review. **Alternative Flow 2: Process Exceptional Activity Appeal**

	1. The Student submits an appeal concerning an exceptional activity outside the system. 2. The system routes the appeal to the Faculty-level Assessment Officer. 3. The Faculty Assessment Officer reviews the Student's justification and supporting evidence. 4. The Officer verifies the information according to the applicable training point regulations. 5. The Officer approves or rejects the appeal and provides a reason for the decision. 6. The system updates the relevant record, records the decision in the audit trail, and notifies the Student.
Pre-conditions	A Student has submitted an appeal related to an activity managed by the Organization. The appeal is within the permitted submission period and has not exceeded the appeal limit.
Post-conditions	The appeal is resolved by the responsible Organization. The decision and its corresponding reasons are recorded, and the Student is notified.

2.10 Use-case UC10: System Audit Trail

Use case Name	System Audit Trail.
Brief description	The Administrator to view and audit system logs, including critical actions related to account permissions, regulation configurations, evidence verification, activity moderation, and training point assessment decisions.
Actors	Administrator.
Basic Flow	1. The Administrator navigates to the System Audit Trail section. 2. The system displays a chronological log of critical actions, including: - Account creation, permission granting, and permission revocation. - Training point regulation creation, modification, and version assignment. - Evidence approval, rejection, and additional evidence requests. - Activity approval, rejection, and cancellation. - Exceptional record verification and point assignment. - Faculty-level score proposals and score adjustments. - University-level Official Score approval and adjustments. 3. The Administrator applies filters (by user ID, action type, date). 4. The system displays the filtered results showing who performed the action, when, what action and the recorded reasons for score adjustments.
Alternative Flows	**Alternative flow 1: Export logs** 1. At step 4, the Administrator clicks "Export". 2. The system generates and downloads a secure log report file.
Pre-conditions	The Administrator is logged in with valid privileges.
Post-conditions	The Administrator successfully audits system activities for accountability.

2.11 Use-case UC11: Handle Exception Cases

Use case Name	Handle Exception Cases.
Brief description	An Faculty-level Assessor verifies and awards points for off-system evidence (e.g., Olympic competitions, external blood donation).

Actors	Faculty-level Assessment Officer.
Basic Flow	1. The Faculty Assessment Officer selects a pending exceptional record from the dashboard. 2. The system displays the Student's submitted evidence and related activity information. 3. The Faculty Assessment Officer reviews the uploaded certificates or supporting documents. 4. The Faculty Assessment Officer verifies the authenticity and eligibility of the record according to the applicable training point regulations. 5. The Faculty Assessment Officer assigns the appropriate criteria and points. 6. The system updates the record status to "VALID" and includes the awarded points in the Student's Predicted Score.
Alternative Flows	**Alternative flow 1: Request revision** 1. At step 3, if the evidence is insufficient, the Assessor clicks "Return for Revision". 2. The Assessor enters the missing requirements and the system notifies the Student. 3. The system updates the record status to "REQUIRES ADDITIONAL EVIDENCE". 4. The system notifies the Student. 5. The Student submits the revised evidence. 6. The system updates the record status to "PENDING" and returns it to the Faculty Assessment Officer for further review. **Alternative Flow 2: Reject Exceptional Record** 1. At step 4 of the basic flow, if the submitted evidence is invalid or does not meet the applicable regulations, the Faculty Assessment Officer rejects the record. 2. The Officer enters a rejection reason. 3. The system updates the record status to "INVALID" and notifies the Student.
Pre-conditions	A Student has submitted evidence for an exceptional activity outside the system, and the record is pending review by the Faculty Assessment Officer.
Post-conditions	External evidence is processed and integrated into the student's Predicted Score.

2.12 Use-case UC12: Aggregate Points & Propose (Faculty)

Use case Name	Aggregate Points & Propose (Faculty).
Brief description	A Faculty Assessor reviews Predicted Scores, resolves discrepancies, and proposes final results to the University Assessor.
Actors	Faculty Assessor.
Basic Flow	1. The Faculty Assessor opens the point aggregation dashboard for their faculty. 2. The system displays students' Predicted Scores, evidence statuses, exceptional records, and applied maximum limits by criteria. 3. The Faculty Assessment Officer reviews each student's assessment records, including:

	- Points accumulated under each training point criterion. - Evidence statuses and supporting documents. - Verified exceptional records and their assigned points. - Any discrepancies or missing information requiring review. 4. The Faculty Assessment Officer selects "Propose Results" after completing the review. 5. The system updates the status to "FACULTY RECOMMENDED". 6. The system forwards the proposals to the University Assessor.
Alternative Flows	**Alternative flow 1: Score adjustment during proposal** 1. At step 3 of the basic flow, if the Faculty Assessment Officer identifies a discrepancy, the Officer adjusts the proposed score within their authority. 2. The system requires the Officer to provide a documented reason for the adjustment. 3. The system saves the adjustment and records the action in the audit trail. 4. The Officer resumes the review and proceeds to propose the results.
Pre-conditions	It is the end of the evaluation period. Student assessment records and relevant supporting evidence are available for faculty-level review.
Post-conditions	Faculty-level assessment proposals are finalized, recorded with supporting information and adjustment reasons when applicable, and forwarded to the University Assessment Officer with the status "FACULTY RECOMMENDED".

2.13 Use-case UC13: Moderate Activities

Use case Name	Moderate Activities.
Brief description	Assessment Officers to review and approve activities proposed by Organizations within their respective management levels before publication. Faculty-level Assessment Officers review faculty activities, while University-level Assessment Officers review university-level activities.
Actors	Assessor (Faculty or University).
Basic Flow	1. The Assessor views the pending activities list. 2. The Assessor clicks on an activity to review its details and proposed criteria. The system displays the activity details and proposed criteria based on the Organization's management level. The responsible Assessment Officer reviews the activity within their assigned scope. 3. The Assessor clicks "Approve". 4. The system updates the activity status to "Approved" (Open/In Progress). 5. The system publishes the activity to students.
Alternative Flows	**Alternative flow 1: Reject proposed activity** 1. At step 3, if the activity details violate regulations, the Assessor clicks "Reject". - The Assessor enters a rejection reason. - The system records the rejection reason in the audit trail. - The system notifies the corresponding Organization. 2. The Assessor inputs a reason, and the system returns the activity to the Organizing Unit as a "Draft".
Pre-conditions	An Organization has submitted an activity with "Pending Approval" status. The activity is assigned to the Assessment Officer responsible for its corresponding management level.

Post-conditions	The activity is either published to students or returned for modification.

2.14 Use-case UC14: Manage Rule Configurations

Use case Name	Manage Rule Configurations.
Brief description	The Administrator creates and manages versions of training point regulations and assigns them to academic semesters.
Actors	Administrator.
Basic Flow	1. The Administrator navigates to the Regulation Configuration module. 2. The Administrator clicks "Create New Version". 3. The Administrator configures the training point criteria, weights, maximum score limits, and addition/subtraction rules. 4. The Administrator assigns the regulation version to a specific semester or academic year and defines its effective period. 5. The system validates the configuration to ensure that the assigned effective period does not conflict with other applicable regulation versions. The Administrator saves the configuration. 6. The system activates the new regulation version for the designated period while preserving previous versions and their associated assessment records.
Alternative Flows	**Alternative Flow 1: Attempt to Modify Previous or Conflicting Rules** 1. At step 5 of the basic flow, if the configuration overlaps with a finalized semester or attempts to overwrite a previous regulation version, the system blocks the action. 2. The system displays an error message and prompts the Administrator to select a valid effective period or create a new regulation version. 3. The previous regulation versions and their associated historical assessment records remain unchanged.
Pre-conditions	The Administrator is logged in with valid regulation configuration permissions. The applicable training point regulations and assessment criteria are available for configuration.
Post-conditions	A new regulation version is saved and assigned to its designated semester or academic year. Previous regulation versions and finalized historical assessment records remain unchanged.

2.15 Use-case UC15: Manage Accounts & Permissions

Use case Name	Manage Accounts & Permissions.
Brief description	The Administrator verifies identities and grants access permissions to sensitive accounts (Organizations, Assessors).
Actors	Administrator.
Basic Flow	1. The Administrator accesses the Account Management dashboard. 2. The Administrator reviews the list of pending sensitive account requests. 3. The Administrator verifies the identity of the requester outside the system. 4. The Administrator selects the account, assigns the corresponding role (Faculty Org, University Assessor, etc.), and clicks "Approve". 5. The system logs the authorization action in the audit trail.
Alternative Flows	**Alternative flow 1: Revoke permissions** 1. The Administrator selects an existing Organization or Assessor account. 2. The Administrator clicks "Revoke Role" and inputs a reason.

	3. The system removes the privileges and logs the action.
Pre-conditions	A user has requested an Organization or Assessor account.
Post-conditions	System access control is securely maintained and logged.

2.16 Use-case UC16: Approve Official Points (University)

Use case Name	Approve Official Points (University).
Brief description	A University Assessor reviews faculty proposals, cross-checks evidence, and approves final Official Scores.
Actors	University Assessor.
Basic Flow	1. The University Assessor selects a specific semester. 2. The system displays the faculty-level proposals, supporting evidence, exceptional records, university-level activity results, and Predicted Scores for review. 3. The Assessor reviews the records and clicks "Approve Official Score". 4. The system logs the approval action in the audit trail. 5. The system updates the status to "UNIVERSITY APPROVED" and finalizes the Official Score.
Alternative Flows	**Alternative flow 1: Return to Faculty for revision** 1. At step 3, if discrepancies are found, the Assessor clicks "Return to Faculty". 2. The Assessor inputs a mandatory reason for return. 3. The system updates the status to "RETURNED TO FACULTY". **Alternative flow 2: Direct score adjustment** 1. At step 3, if the Assessor directly changes a score, the system prompts for a mandatory reason. 2. The Assessor inputs the reason and saves. 3. The system logs the adjustment and finalizes the score.
Pre-conditions	Faculty Assessment Officers have completed their reviews and submitted the proposed training point results with the status "FACULTY RECOMMENDED". The proposals and their supporting records are available for University-level review.
Post-conditions	The assessment is either finalized with the status "UNIVERSITY APPROVED", and the Official Score is locked and announced to the Student, or returned to the Faculty for revision with the status "RETURNED TO FACULTY".